

TEM Classification of Aortic Dissection—The Evolving Scoring System: A Literature Review



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Aortic dissection is an acute presentation that, if unnoticed, poses a significant risk to life. Anatomically, it is defined as a tear in the intimal layer of the aorta, but management differs significantly based on the location of this tear. Traditionally the Stanford and DeBakey classifications have been used to distinguish tear types and thus guide the most favourable management option, be it medical optimisation or surgery. Recently, a new Type-Entry-Malperfusion classification has been proposed to more accurately define and thus risk stratify patients with aortic dissection. This review summarises the Type-Entry-Malperfusion classification and highlights its potential advantages and limitations compared to other classifications. Clinical insights and potential barriers to adopting this classification are also described in this review.

Keywords

Aortic Dissection • Classification • Tear • Stanford classification

Introduction

Acute aortic dissection (AAD) is a critical and potentially fatal condition characterised by an intimal tear in the ascending aorta. The tear leads to blood flow between the aortic intima and media resulting in the formation of a false lumen within the aortic wall. This false lumen can propagate down the aorta and cause a range of complications including organ malperfusion and potentially a catastrophic aortic rupture [1]. Classification systems seek to both describe the

pathology and guide clinical decision-making, determining appropriate treatment strategies. The two primary classification systems used for Type A aortic dissection are the Stanford classification and the DeBakey classification (Figure 1) [2,3]. The Stanford classification categorises dissections based on the extent of the dissection; Stanford Type A dissections involve the ascending aorta, whereas Type B dissections are limited to the descending aorta, distal to the left subclavian artery. The DeBakey classification further subclassifies dissections based on anatomical involvement.

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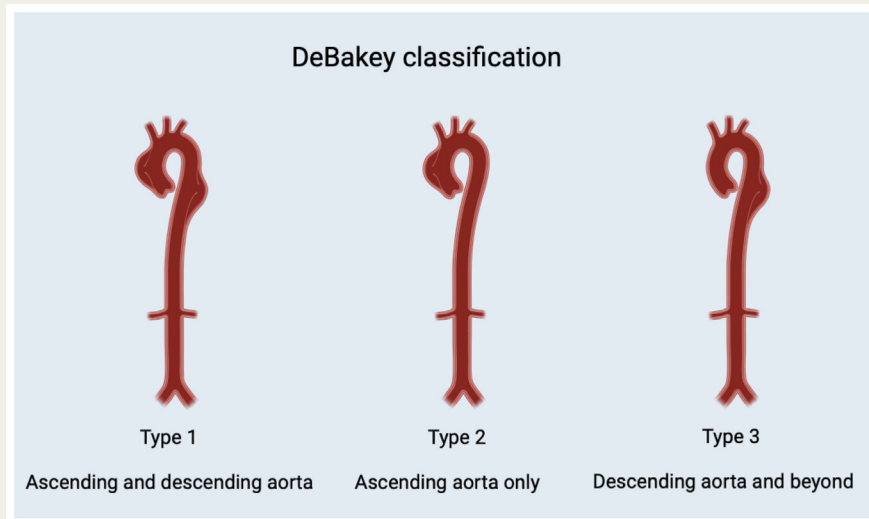


Figure 1 DeBakey classification for aortic dissection.

Management of AAD depends largely on dissection extension, location of the primary entry tear and whether or not there is evidence of organ malperfusion [4]. A recently proposed classification system for aortic dissection incorporates the type (based on the Stanford classification), entry tear location, and malperfusion status (Type-Entry-Malperfusion [TEM]) as its key parameters. In doing so, it introduces a novel framework, including the anatomical character of the lesion as well as key prognostic clinical features. In terms of dissection ‘Type,’ the TEM classification system expands on the Stanford classification by recognising those dissections which arise in the aortic arch but do not involve the ascending aorta; these are described as “non-A-non-B,” and make up approximately 10% of all aortic dissections [4]. By integrating radiological and clinical factors, the TEM classification seeks to comprehensively describe the clinical picture and provide a framework to guide clinical decision-making reliably and consistently.

Evidence Base For TEM Classification

The TEM classification, [Figure 2](#), describes AAD by extension of dissection, site of entry tear, and presence of malperfusion. Type A and B are described as per the Stanford classification. Furthermore, type non-A non-B dissections involve the aortic arch and the descending aorta but not the ascending aorta [4].

In terms of entry tear, they are classified as follows:

- E0 no entry tear identified.
- E1 entry tear in the ascending aorta between the aortic valve and proximal to the brachiocephalic trunk.
- E2 entry tear in the aortic arch between the brachiocephalic trunk and the left subclavian artery.

- E3 entry in the descending aorta, distal to the left subclavian artery.

Finally, malperfusion states are classified as follows:

- M0 no radiological or clinical evidence of malperfusion.
- M1 cardiac malperfusion; dissection of at least one main coronary artery, with (M1+) or without (M1–) evidence of ischaemia such as ECG changes and/or raised serum cardiac enzymes.
- M2 supra-aortic malperfusion; dissection of at least one supra-aortic vessel or aortic arch true lumen collapse, with (M2+) or without (M2–) clinical symptoms of cerebral (stroke) or upper extremity (pulse deficit, pain, pallor, paraesthesia) malperfusion.
- M3 visceral or lower extremity malperfusion; dissection or false lumen origin of at least one visceral, renal or one iliac artery, or aortic true lumen collapse entailing functional closure of at least one visceral, renal or iliac artery offspring, with (M3+) or without (M3–) clinical symptoms of bowel (abdominal pain, ileus, bloody diarrhoea), kidney (anuria, renal failure signs) or lower extremity (pulse deficit, pain, pallor, paraesthesia) ischaemia; [Figure 1](#).

In reclassifying aortic dissection, Sievers *et al.* [4] reported that 10% of their patients presented with non-A non-B dissection. The identification of this group of patients who do not fit clearly into either the Stanford or DeBakey classifications supports the introduction of an updated classification system. Interestingly, patients presenting with non-A non-B dissections were significantly younger than those presenting with Type A or Type B (59 years vs 65 years vs 67 years, $p=0.01$). Unsurprisingly, they found mortality to be higher in the Type A group than in types non-A non-B and Type B (16% vs 8% vs 5%, $p=0.01$). This trend was also observed in the long-term survival figures which demonstrated 2-year

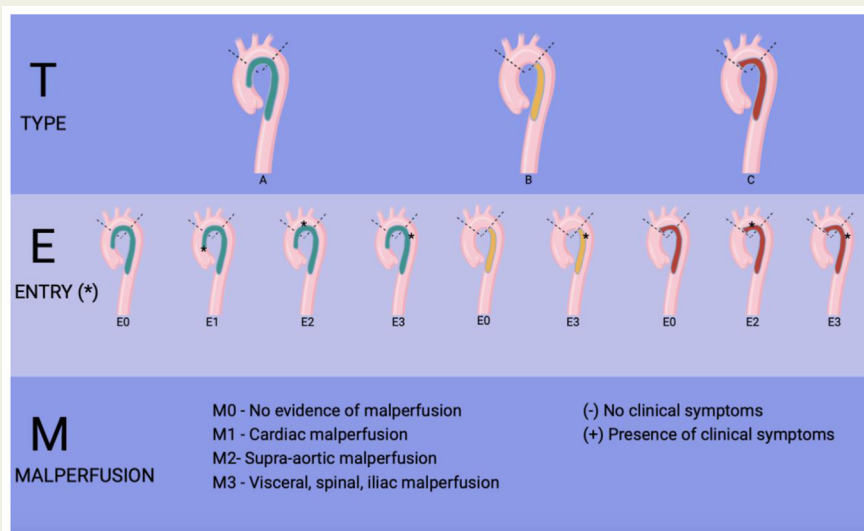


Figure 2 TEM (type/entry location/malperfusion) classification for acute aortic dissection.

survival to be lowest among patients with Type A dissection, followed by non-A non-B and Type B ($83 \pm 3\%$ vs $88 \pm 6\%$ vs $93 \pm 3\%$, $p=0.019$). Within the Type A group, the highest risk patients were those classified as “E2,” with an entry tear in the arch. Interestingly, patients with non-A non-B dissection had the highest rate of aortic re-intervention compared to Type A and Type B (25% vs 12% vs 21% , $p=0.07$) [4].

Wang et al. [5] applied the TEM classification to patients presenting with AAD at their institution between January 2019 and May 2020. Of the 151 patients they identified, 19.2% presented as non-A non-B dissections, confirming this as a significant cohort of patients. They noted that patients with an entry tear in the arch (E2) had the highest rates of neurological complications compared to E0, E1, and E3 (41.4% vs 8.3% vs 20.0% vs 16.7% , $p<0.05$). Malperfusion of the heart (M1) was associated with high rates of mortality, therefore they recommended emergency surgery in patients with Type A E1 M1. They also described different distributions of procedures performed according to the TEM classification. Patients with Type A all underwent ascending aorta replacement; those with an entry tear in the arch or descending aorta (E2 or E3) were significantly more likely to also undergo total arch replacement than those with E1. Patients with supra-aortic malperfusion (M2) were likely to undergo total arch replacement. Crucially, they demonstrated in a different patient population the utility of the TEM scoring system and recommended its ongoing use as a supplement to the Stanford classification.

History of TEM Classification

There has been a leap in exploring and actively understanding the natural course of aortic dissections during the past two decades. Consequently, this has enabled an improvement in

treatment options and advantageous outcomes. The two most used classification systems for aortic dissections are the Stanford and DeBakey systems, introduced in the 1950s and 1960s respectively [6]. Firstly, the Stanford classification system relies on the extent of the dissection process irrespective of the location of the primary tear. The Stanford Type A dissection includes any part of the proximal aorta to the origin of the left subclavian artery, while Type B is limited to the descending aorta, distal to the left subclavian artery origin [7]. Secondly, the DeBakey system classifies aortic dissections anatomically and similarly focuses on ascending and descending involvement in the dissection and the primary entry tear site [8]. Furthermore, since this is a potentially fatal condition, a fast diagnosis is crucially required to improve patient outcomes and chances of survival. Diagnostic imaging modalities such as computed tomographic angiography (CTA), magnetic resonance angiography, and sonography play a vital role in diagnosis accuracy and follow-up.

The treatment of AAD heavily relies on the dissection's extension, primary entry tear location, and organ malperfusion [4]. An accurate and precise classification system was urgently needed that combines both imaging modalities and interpretation as well as treatment. In 2019, Sievers et al. [4] proposed a novel classification system for aortic dissection based on a modified Stanford classification, that incorporates TEM). The most vital component considered as the primary aim of therapy for AAD is the exclusion of the primary tear from the circulation (whether that be through surgical resection or different modalities such as transcatheter procedures), which determines the presence or absence of malperfusion [9]. For this reason, the TEM classification was introduced to offer a more accurate and precise way to evaluate disease complexity and classify patients presenting with AAD. This system aids in

awareness of the disease mechanism, enhances decision-making regarding the extent of repair, and helps in anticipating outcomes.

TEM Score Compared With Other Widely Used Scoring Systems

In recent years, there have been huge advances in the investigation and management of aortic dissection, as well as in the understanding of aortic pathologies. However, despite such advances, regarding classifying aortic dissection, the Stanford classification continues to be the system of choice [4]. Notably, this system was created prior to the development of endovascular treatments and before aortic imaging was widely available [4]. The mainstay of the Stanford system was to determine whether a patient would benefit from medical or surgical management [4]. However, given the introduction of new surgical techniques, such as endovascular repair, hybrid aortic arch replacement, and the frozen elephant trunk technique, as well as advancements in imaging, it is evident that a system developed in 1970 would benefit from some modernisation [4]. Moreover, the Stanford classification system does not take into account all anatomical variants possible within aortic dissections, rather simply categorising dissections into Type A or Type B, if they concern the ascending or descending aorta respectively [4]. This has resulted in a degree of ambiguity regarding the treatment of dissections that involve the aortic arch, despite this variant comprising 10% of all dissections [4]. Such dissections may occur due to an intimal tear within the descending aorta that then expands into the arch, or simply as a tear that occurs within the aortic arch itself [4]. Previous studies have highlighted the shortcomings of the Type A, Type B classification system; Von Segesser *et al.* [10] suggested the use of a “non-A non-B” category to encompass dissections involving the arch. There was a high degree of variability in the frequency of “non-A non-B” dissections reported in the literature, ranging from 2% to 11%, this is likely due to the need for further clarity regarding the standardisation of this category [10–14]. Additionally, the uncertainty surrounding this category has led to discrepancies in the management of “non-A non B” dissections; a number of studies have recommended surgical management of such cases and have demonstrated increased mortality rates associated with conservative management [11]. While the management, of course, depends on the location of the dissection, the patient outcome and appropriate treatments also vary significantly depending on the presence and degree of malperfusion, a variable that many traditional classification systems fail to take into account [4]. Patients suffering from malperfusion are at risk of further complications, including acute renal failure, and frequently require more intensive treatments, such as endovascular therapies like stent graft replacement [11]. There have been previous attempts to involve the degree of ischaemia within the classification; the Penn system, for example, incorporates

malperfusion status into the Stanford system [11]. However, this system was reserved solely for Type A dissections and failed to take into account the Type of malperfusion present [3]. Surgeons have also been reluctant to accept the Penn system, with new terminology viewed by some as impractical and inconvenient [4].

The TEM classification meets many of the shortcomings of the aforementioned systems; it recognises “non-A non-B” as a unique entity, thus removing any doubt surrounding such patients, and aiding rapid decision-making within the emergency department [4]. It is imperative to perform a thorough diagnostic examination when using the TEM system, this will further aid the relevant surgical team to provide the patient with the most appropriate management [4]. This system has highlighted particularly high-risk anatomical variants, for example, the highest mortality occurred among patients who suffered a Type A dissection with an entry location within the aortic arch, awareness of the risks with such variants may aid clinicians in making decisions regarding management and thus enhance survival [4]. Furthermore, it encourages clinicians to consider whether patients will benefit from more aggressive treatment, as it takes into account malperfusion status and further allows accurate prediction of in-hospital mortality [4]. The system is succinct, straightforward to use and should be beneficial for both emergency physicians and the relevant surgical teams [4].

Advantages

When considering the uses of the “TEM” classification system when diagnosing and managing patients with aortic dissection, one of the main advantages as elucidated by Sievers *et al.* [4] is the breadth of parameters considered when classifying aortic dissection patients, that supersede the current classification systems. The expansion in the type of dissection to include non-A/non-B dissections alone allows for further sub-categorisation, analysis of patients and exploration of management options. The addition of a location for primary entry tear identified by modalities such as CTA aids in pre-surgical planning. This is pivotal as aortic operations have advanced so that every aortic segment is replaceable either open or endovascularly. Furthermore, the inclusion of malperfusion into the classification system helps clinicians to stratify patients in light of clinical urgency [11]. Coronary malperfusion was observed only in patients with Type A dissection and visceral or iliac malperfusion was observed in all main categories of dissection [4].

The “TEM” system sits at precedence when predicting the likelihood of end-organ failure and mortality outcomes in a larger cohort of patients who were previously in a grey area, unable to meet classification criteria. This offers a more precise description of aortic dissection that can be used at the first point of contact in secondary care settings and allows clinicians to strategically prepare for potential complications associated with the dissection presented.

How Can the TEM Score Be Improved

Aortic dissection is an emergency, requiring prompt treatment with the combined effort of emergency physicians, cardiac surgeons, and radiologists. The popularity of the Stanford and DeBakey classifications can be attributed to their simplicity when describing the pathology, thus removing ambiguity in the management of AAD. While the TEM system adds further anatomical detail to the AAD, the added letters and numbers may be confusing in a high-stress situation, which may be a barrier to the adoption of this system in clinical practice. Additionally, the extra detail provided by the TEM system is usually conveyed verbally from the radiologist to the surgeons, suggesting a limited prognostic benefit from the TEM system. In contrast, it can be argued that the purpose of the TEM system was to provide additional information, which can be used alongside the Stanford classification, at a later chronological point in the management of AAD. While it is successful in doing so, it can be argued that there is still missing detail within this system. For example, M3 would suggest malperfusion in either a visceral, renal, or iliac artery, however, it does not differentiate between them. Additionally, the location of these three different vessels is vastly different, thus there is significant heterogeneity as to what M3 defines. Therefore, the TEM system would benefit from subclassification of the M section to reduce ambiguity, but this may overcomplicate the system.

In addition to providing clarity for decision-making in AAD, anticipating the outcome of surgery is another purpose of the TEM system. Another widely used scoring system for the prediction of mortality post-cardiac surgery is the European System for Cardiac Operative Risk Evaluation II (EuroSCORE II) [15]. EuroSCORE II includes the presence of comorbidities such as chronic pulmonary dysfunction and diabetes to indicate a worse mortality post-surgery, which the TEM system can benefit from. Additionally, the TEM system does not include preoperative ejection fraction, a variable that significantly impacts mortality. Lin et al. [16] demonstrated a 68% increased mortality among those with a low ejection fraction, suggesting that preoperative ejection fraction may influence suitability for surgery.

The simplicity of the Stanford classification allows for efficient decision-making and reduced ambiguity. Therefore, the use of the TEM system as an adjunct to the Stanford classification can synergistically provide additional detail while reducing ambiguity when communicating across specialties. If used as an adjunct, the TEM system may also benefit from the inclusion of ejection fraction and the presence of comorbidities which can provide robustness to the system. The TEM system highlights that a gap in the management of non-A non-B AAD has been identified and, once the barriers to its implementation have been overcome, it has the potential to provide prognostic benefit to the management of AAD.

Limitations

When discussing the TEM system, it is imperative to note that it is not always possible to detect entry tear locations with CTA. While the system does provide a way of discussing AAD in more detail, there is currently a lack of data regarding the implications of different subcategories on operative timing. Further prospective studies looking at this will help add to its real-world use in influencing operative timing, particularly in non-A non-B groups. It also remains to be seen whether the classification performs consistently well across different patient cohorts, for example, those with connective tissue disorders, as well as across different age ranges.

The system places an emphasis on the poor prognostic implications of entry tear location. The original study was able to confirm the entry tear location in Type A dissections intraoperatively, however, this was not possible in patients who suffered non-A, non-B or Type B dissections who were managed conservatively or who received endovascular treatment. As such, reports regarding tear entry location in such patients are subject to false negative results [4]. Furthermore, due to limited patient numbers, the original study was not able to analyse patient outcomes within all subcategories [4].

Conclusion

The classification of AAD is heterogeneous and certain patients present with non-A non-B dissections. The TEM system adds prognostic detail in this cohort of patients and provides clarity on whether and what type of surgical repair is required. In the future, the adoption of the TEM system can provide nationwide standardisation of non-A non-B dissections as well as AAD. Current barriers to its implementation include the complexity and the difficulty of adopting new terminology in a well-known condition.

Conflicts of Interest

There are no conflicts of interest to disclose.

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